MIL-S-43915A(GL)
28 September 1977
SUPERSEDING
MIL-S-43915(GL)
27 December 1974

MILITARY SPECIFICATION

SHELTER, EXPANDABLE FOR MEDICAL UNIT
SELF-CONTAINED, TRANSPORTABLE (MUST)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a rigid, expandable shelter constructed of aluminum-faced, paper honeycomb core sandwich panels and having the following approximate dimensions (folded): length 12 feet 11 inches, width 7 feet 8 inches, height 8 feet, (3.94 m by 2.33 m by 2.43 m). The approximate dimensions, expanded, are as follows: length 19 feet 3 inches, width 12 feet 11 inches, height 8 feet, (5.89 m by 3.94 m by 2.43 m).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5410

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:57Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

QQ-P-416	-	Plating, Cadmium (Electrodeposited)
TT-E-527	-	Enamel, Alkyd, Lusterless
TT-E-529	-	Enamel, Alkyd, Semigloss
TT-P-1757	-	Primer Coating, Zinc Chromate, Low Moisture Sensitivity
MMM-A-132	-	Adhesives, Heat Resistant, Airframe Structural, Metal to Metal

MILITARY

MIL-B-5087	-	Bonding, Electrical, and Lighting Protection, for Aerospace Systems
MIL-W-5088	-	Wiring, Aircraft, Installation of
MIL-C-5541	-	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-A-25463	-	Adhesive, Film Form, Metallic Structural, Sandwich Construction
MIL-C-26482	-	Connector, Electric, Circular, Miniature, Quick Disconnect, Environment Resisting
MIL-A-43759	-	Adapter, Air Lock, MUST
MIL-L-43768	-	Loading Kit, MUST Shelter
MIL-A-43820	-	Airlock, Medical Unit, Self-Contained, Transportable (MUST)
MIL-T-43821	-	Tie-Down Assembly; Cargo (MUST)
MIL-B-43909	-	Bellows, Expandable Shelter (MUST)
MIL-A-83377	-	Adhesive Bonding, (Structural) for Aerospace Systems, Requirements for

STANDARDS

FEDERAL

FED-STD-595	-	Colors

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Property
MIL-STD-401	-	Sandwich Constructions and Core Materials; General Test Methods
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of
MS-63045	-	Plate, Identification, Miscellaneous

Source: https://assist.dla.mil -- Downloaded: 20260-21T22:57Z
Check the source to verify that this is the current version before use.

DRAWINGS

ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

5-4-2000 - Shelter, Expandable, for Medical Unit, Self-Contained, Transportable (MUST); Drawing List

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code

(Application for copies of NFPA Standards, should be addressed to the National Fire Protection Association, 60 Batterymarch Street, Boston, MA 02110.)

3. REQUIREMENTS

* 3.1 Panel process and sample approval. Prior to award of contract, the contractor shall submit to the contracting officer a certified copy of a laboratory test report and a copy of the process specification covering fabrication of the metallic-faced, paper honeycomb core sandwich panels he proposes to use in the construction of the end item. Using his process specification, the contractor shall fabricate a qualification test sandwich panel and shall perform the tests for sandwich panels specified in 4.4.1.1.3. The contractor's process specifications shall contain an adequate description for the control of materials, storage conditions prior to use, cleaning of materials, priming or otherwise protecting aluminum skins from corrosion, fabrication, quality standards and control which provide for the surveillance, inspection and testing he proposes to use to manufacture the panels and which will assure continued quality of product. A sample specimen of the test panel 24 inches by 24 inches (610 by 610 mm) minimum, fabricated using 0.040 inch (1.0 mm) aluminum skins shall be submitted along with the test report and process specification to the contracting officer or his authorized representative for inspection. The laboratory test report shall contain test data which demonstrates that the panels he proposes to use have been tested and found to comply with all the requirements specified herein. Detailed guidance for the preparation of and requirements to be included in the contractor's process specification is provided in MIL-A-83377. The contractor shall submit to the contracting officer the identification of the adhesive he intends to use and certification that this adhesive is capable of meeting all requirements of MMM-A-132, type 1, class 2, except metal to metal lap shear strengths shall be as follows:

(a) Metal to metal structural bonds, adhesive shear

Temperature	Shear strength
-65°F (-53.9°C)	3000 psi (20.7 MPa)
Ambient	3000 psi (20.7 MPa)
200°F (93.3°C)	2000 psi (13.8 MPa)

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:57Z
Check the source to verify that this is the current version before use.

(b) Humidity exposure - The adhesive shall withstand exposure to 100 percent relative humidity at 200°F (93.3°C) for two weeks and meet a 1500 psi (10.3 MPa) minimum shear strength at 200°F (93.3°C)

(c) Salt spray exposure - The minimum strength of the adhesive shall be 2000 psi (13.8 MPa) after exposure to salt spray (20 percent NACl solution) at 95°F (35°C) for two weeks.

Within thirty days after award of contract, the contractor shall submit to the contracting officer certified proof that the above tests (a, b and c above) have been accomplished and satisfactory results obtained (see 6.7).

3.2 Alternate components. Components offered as equivalent to components specified hereinafter and on the referenced drawings as specific manufacturer's part number or equal shall be functionally equal, of equal or better quality, with the manufacturer's part number identified. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and shall not reduce ease of maintenance to it or any other components, unless such modification or change is specifically approved by the contracting officer. Prior to manufacture of the first article, or if none is required, prior to commencing production, the contractor shall submit for the contracting officer's approval, a list identifying each proposed alternative component, together with proof that each listed component complies with requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of the first article or production manufacture as applicable and does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.3 First article. When specified (see 6.2), the contractor shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.3 and 6.3).

3.3.1 Initial production testing sample. When specified (see 6.2), the contractor shall furnish one or more shelters for initial production inspection and approval (see 4.3.1 and 6.3.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:57Z
Check the source to verify that this is the current version before use.

3.3.2 Inspection comparison testing sample. When specified (see 6.2), the contractor shall furnish a sample for comparison inspection and approval (see 4.3.2 and 6.3.2).

3.4 Materials and components. Materials and components shall conform to the specifications, standards and drawings listed in section 2 and as specified herein. When materials are not definitely specified, they shall be of the quality normally used by the manufacturer of hard shelters provided the completed item complies with all provisions of this specification (see 6.8).

3.4.1 Sandwich panels. Sandwich panels shall be hot bonded using aluminum face sheets, paper honeycomb core and foam materials specified on drawings. Thicknesses of finished panels and core ribbon directions shall be as specified on the drawings. Where core splicings occur the panel shall maintain structural soundness and shall be made either by over lapping or interlocking the honeycomb core segments to be spliced not less than 1 inch (25 mm) nor more than 2 inches (51 mm) or by butting core segments and/or faying surfaces and bonding the surfaces over the entire length of the splice. Where core splicing is accomplished by the overlapping or interlocking of core cells (resulting in crushing of cells) a foaming-type adhesive shall be used to completely fill the entire splice area to prevent any migration of water through the splice. Face sheets, core and edging shall be integrally bonded with adhesive. The fully cured bond shall be capable of meeting the physical requirements specified herein.

3.4.1.1 Physical properties. Finished panels shall exhibit no surface defects such as bulges, depressions, dents and scratches. Unless otherwise specified on the drawings, tolerance for panel warpage shall be as follows:

Warpage - plus or minus 0.025 inch (0.66 mm) per foot (305 mm) length of panel

The finished panels shall be resistant to fire, humidity and age and shall conform to the following requirements when tested as specified in 4.4.1.1.3:

Fire resistance - Nonburning, self-extinguishing within 15 seconds after removal of the flame with no smoldering or glowing visible for a maximum of 10 seconds thereafter.

Humidity and age - No warping, staining, delamination or other adverse effect on appearance or function.

4 Lb (1.8 kg) Core

Face sheet to core bond:

Peel strength (min) - 4 inch-pounds (0.45 N·m) of torque per inch (25 mm) of width

Shear beam strength (min):

TL plane - 175 psi (1.2 MPa)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:57Z
Check the source to verify that this is the current version before use.

TW plane - 105 psi (0.72 MPa)

Tension strength - 400 psi (2.76 MPa) or core failure

Compressive strength - 375 psi (2.59 MPa)

Continuity of bond - no void area

2.5 Lb (1.1 kg) Core

Face sheet to core bond:

Peel strength (min) - 4 inch-pounds (0.45 N·m) of torque per inch (25 mm) of width

Shear beam strength (min):

TL plane - 75 psi (517 kPa)

TW plane - 36 psi (248 kPa)

Tension strength - 200 psi (1.4 MPa) or core failure

Compressive strength - 165 psi (1.1 MPa)

Continuity of bond - no void area

3.4.1.2 Panel identification. Each sandwich panel shall be permanently and legibly marked for identification with a serial number or part number, the lot number, and the date of manufacture.

3.5 Design and construction. Design and construction of the expandable shelter shall conform to the drawings listed on Drawing 5-4-2000 and as specified hereinafter. The unit shall be free of panel delaminations and shall meet all physical and environmental requirements specified herein.

3.5.1 Leveling, opening, and setting up. There shall be no difficulty in leveling, opening and setting up the unit (see 3.5.5).

3.5.2 Internal air leakage. Internal air leakage from the unit shall not exceed 76 standard cubic feet per minute (scfm) (0.04 m^3/s) when the unit is in its open configuration and pressurized to 0.3 inch of water (water gauge) (74.7 Pa), (see 4.6.6).

3.5.3 Blackout. There shall be no internal light visible from outside of the unit in either the open or closed configuration when tested as specified in 4.6.7.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:57Z
Check the source to verify that this is the current version before use.

3.5.4 Structural soundness. The unit shall maintain structural soundness and shall show no evidence of delamination, deformation or damage, loosening or weakening of structural parts when subjected to the following:

a. When fully loaded (5500 pound (2.5 Mg) dummy load):

1. Towed on its skids (see 4.6.8).

2. Transported on a demountable dolly set (see 6.5).

3. Subjected to railroad humping (see 6.5).

b. When loaded with a dummy load to a gross weight of 25,000 pounds (11.3 Mg) (unit and dummy load) and hoisted by the lifting rings (see 4.6.11).

c. When in its folded configuration with another unit, or equivalent, stacked on its top (see 4.6.12).

3.5.5 Environmental characteristics.

3.5.5.1 Water leakage. The shelter in its closed and erected configurations shall show no evidence of water leakage, penetration, or absorption, when subjected to a simulated rainfall of 4 ± 1 inches (102 ± 25 mm) per hour as specified in 4.6.13.

3.5.5.2 Low temperature. There shall be no delamination of panel surfaces or separations of edge extrusions or permanent warping when the unit is subjected to a low temperature of minus 65° ± 5°F (-53.9 ± 3°C) for 24 hours (see 4.6.14).

3.5.5.3 Humidity. There shall be no evidence of moisture penetration when the unit in its closed configuration, is subjected to a relative humidity of 97 ± 2 percent (see 4.6.15).

3.5.5.4 Atmospheric pressure. There shall be no evidence of damage to the panels when subjected to an atmospheric pressure of 87 mm of mercury (11.7 kPa) (corresponding to that pressure encountered at 50,000 feet (15.2 km) above sea level) (see 4.6.16).

3.5.5.5 Solar load. The unit shall show no evidence of delaminations, buckling or other deterioration or permanent deformation when subjected to simulated solar loads sufficient to raise the outer skin temperature to 200° ± 5°F (93.3 ± 3°C) (see 4.6.17).

3.5.5.6 Heat transfer. The overall coefficient of heat transfer shall be no greater than 0.23 British thermal units (Btu's) per hour per degree Fahrenheit per square foot (1.3 $W/m^2 \cdot K$) of outside shelter area (see 4.6.18).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:57Z
Check the source to verify that this is the current version before use.

* 3.5.6 Installation of closeouts, inserts. Installation of edge extrusions, closeouts and inserts shall be as shown on the drawings and as specified herein. Aluminum alloy edge extrusions shall be cleaned and prepared for bonding in accordance with the cleaning, preparation and handling methods specified for the aluminum alloy face sheets used in the manufacture of the sandwich panels. Phenolic edging shall be cleaned, on the side to be bonded, by a light sanding to remove foreign matter and to roughen the surface for bonding. Prior to assembly of edge members, closeouts and inserts to the sandwich panels, sufficient adhesive shall be applied to insure complete protection on honeycomb paper core from moisture penetration and to insure total contact with edge members and inserts to be applied. Both sides of the aluminum edge extrusion leg, the faying surfaces of the extrusions and panel skins shall be coated with the adhesive specified on the drawings. Use of the adhesive shall be in accordance with the adhesive manufacturer's instructions for application, curing, pot life and age. No gaps shall be visible between aluminum faces and extruded or phenolic edges or closeouts and at the intersection of the extruded edges or closeouts.

* 3.5.6.1 Inserts. All inserts shall be in accordance with Drawing 5-4-2079. When tested as specified in 4.6.2.2 and 4.6.2.3, the inserts shall withstand the torque and pull-out loads specified in table I without failure of the insert; panel or potting compound.

TABLE I. Insert strength

Insert Thread size, dia	Pull out force (lb)	Pull out force (kN)	Torque (ft-lb)	Torque (N·m)
10-32, 9/16	1000	4.4	10	13.6
1/4-28, 7/16	1000	4.4	20	27.1
5/16-24, 11/16	1500	6.7	25	33.9
3/8-24, 3/4	1500	6.7	50	67.8
5/16-24, 1/2	1000	4.4	20	27.1

3.5.7 Electrical assembly. All interior power-outlet receptacles, switches and telephone jacks shall be located as shown on the applicable drawings. All wiring shall be in accordance with MIL-W-5088. All wiring in raceways with removable covers shall be secured by a clamping means specified in MIL-W-5088. Raceway wiring ducts and junction boxes shall be free from objects such as clips, screws, nuts and rivets and shall be free from burrs or sharp corners which can damage or cause deterioration of insulation of wires or cables.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:57Z
Check the source to verify that this is the currentversion before use.

Connectors and receptacles shall conform to the environmental requirements of MIL-C-26482. Fluorescent light fixtures shall conform to the requirements of applicable drawings and shall meet the requirements for electro-magnetic compatability conforming to class IIb of MIL-STD-461 (see 4.6.19). All exposed metallic parts, wiring raceways and outlet boxes of the electrical distribution system shall be grounded by means of a common ground wire running throughout the shelter. Exposed noncurrent carrying metal parts of the shelter such as the frames, wall panels, removable panels and hinged panels shall be electrically bonded in accordance with Article 250 and Section 501-16 of the National Electrical Code (NFPA #70), and the applicable bonding requirements of MIL-B-5087. All parts of the electrical system shall operate when tested for electrical continuity as specified in 4.6.20.

3.5.8 Plumbing. Plumbing shall be as shown on the applicable drawings. All lines and connections shall be free of leakage when tested as specified in 4.6.21.

3.6 Welding. All welding shall be as specified on the drawings.

3.7 Durability. The complete expandable shelter shall be capable of withstanding a durability test of 1440 hours without failure when tested in accordance with 4.6.24.

3.8 Finish of metal components.

3.8.1 Surface preparation.

3.8.1.1 Steel. All steel components and subassemblies to be painted shall first be clean and free of grease, oil, dirt and other foreign matter and shall be cadmium plated in accordance with type III, class 2 of QQ-P-416.

3.8.1.2 Aluminum. Prior to painting, all aluminum surfaces shall be cleaned and prepared for painting in accordance with class 3 of MIL-C-5541.

3.8.2 Priming. After surface preparation all exposed surfaces, except the interior floor areas, shall be coated with primer conforming to composition G or L of TT-P-1757, color Y or T. The interior floor area shall be primed as specified on Drawing 5-4-2004.

3.8.3 Enameling. When the primer is dry, all exterior surfaces shall be coated with enamel conforming to TT-E-527, olive drab color No. 34087 of FED-STD-595 and all interior surfaces, except for the floor area, shall be coated with enamel conforming to TT-E-529, pale green color No. 24554. The interior floor area shall be finished as specified on Drawing 5-4-2004. Coating shall level out to an adherent, continuous and uniform film without runs, wrinkles, streaks or areas of no film. Any coating damaged during assembly or examination shall be touched up.

3.9 Marking.

3.9.1 Identification. The shelter shall be marked for identification in accordance with MIL-STD-130 and in the locations shown on the drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:57Z
Check the source to verify that this is the current version before use.

3.9.2 Instruction. Instruction markings shall conform to the requirements and in the locations specified on the drawings.

3.9.3 Nameplate. A nameplate, in the size specified and containing the data specified in MS-63045, shall be attached to the unit in the location indicated on applicable drawing.

3.10 Government furnished components.

3.10.1 Adapter, air lock. Each shelter shall be provided with an air lock adapter conforming to type III of MIL-A-43759. When specified (see 6.2), the air lock adapters will be furnished by the Government.

3.10.2 Tie-down assembly, cargo. Each shelter shall be provided with a cargo tie-down assembly conforming to MIL-T-43821. When specified (see 6.2), the tie-down assemblies will be furnished by the Government.

3.10.3 Bellows, expandable shelter. Each shelter shall be provided with a bellows assembly conforming to MIL-B-43909. When specified (see 6.2), the bellows assembly will be furnished by the Government.

3.10.4 Air lock assembly. Each shelter shall be provided with an air lock assembly conforming to MIL-A-43820. When specified (see 6.2), the air lock assembly will be furnished by the Government.

* 3.10.5 External power cables. Each shelter shall be provided with one each external 400 Hz and one each 60 Hz power cable. When specified (see 6.2) the cables will be provided by the Government.

3.11 Government loaned property.

* 3.11.1 Transporting equipment.

3.11.2 Dolly set, lift type. The transportable shelter lift type dolly set will be loaned by the Government (see 3.5.4 and 6.2).

3.11.3 Truck. The truck required for use in towing the dolly set during the transport test shall be a standard 2-1/2 or 5 ton truck and will be loaned by the Government (see 3.5.4 and 6.2).

* 3.12 Workmanship. All metal forming shall be done with dies or fixtures to insure uniformity of size and shape. Sheet metal or structural sections used in fabrication of the shelters shall not be kinked, bent, or have ripples. Rivets, when used, shall be expanded to completely fill the holes, with full-sized heads in full contact with the surface of the member. Bolts, nuts, and screws shall be drawn tight and so secured that they shall not work loose under vibration.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:57Z
Check the source to verify that this is the current version before use.

All surfaces shall be free of burrs, slivers, rough or sharp edges. Parts shall not be fractured, split, punctured, dented or malformed. Parts shall be in visual alignment and securely assembled. All functioning components shall operate as intended without abnormal force. Materials to be welded shall be clean and free from scale, corrosion, oil, water and other matter. Welds shall be sound, smooth, free from flash and shall show good fusion. Welds shall not be cracked and shall be free from slag inclusion, undercut, scale, or flux deposits. Accessories shall be securely fastened and braced to prevent damage or loosening during units' intended use or transportation. The sheet metal in the finished unit shall be as flat as stacked sheet metal. There shall be no areas of rust. Gaskets shall not be torn, cut or split and shall be free of finish. Steel inserts shall be clean and free of dirt, oil, grease or other foreign matter. Excess adhesive shall be removed from joints.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Initial production sample inspection (see 4.3.1).
(c) Inspection comparison (see 4.3.2).
(d) Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required, an expandable shelter shall be examined for the defects listed in table III, dimensions specified, and tested as specified in 4.6.4.5, 4.6.5, 4.6.6, table IV, 4.6.20, 4.6.21, 4.6.22.2 and 4.6.23.2. The presence of any visual defect, any dimension not within the specified requirements or failure to pass any test shall be cause for rejection of the first article.

* 4.3.1 Initial production sample inspection. When specified (see 3.3.1 and 6.2), one or more shelters shall be examined for defects listed in table III, dimensions specified, and tested as specified in 4.6.4.5, 4.6.5, 4.6.6, table IV, 4.6.22.2, 4.6.23.2, and 4.6.24. One complete expandable shelter shall be tested as specified in 4.6.24 and any failure shall be cause for rejection of the initial production sample. The presence of any visual defect, any dimension not within the specified requirements or failure to pass any test shall be cause for rejection of the initial production sample.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:57Z
Check the source to verify that this is the current version before use.

4.3.2 Comparison sample inspection. When a comparison sample is required, the complete shelter(s) shall be examined for defects listed in table III, dimensions specified, and tested as specified in 4.6.4.5, 4.6.5, 4.6.6, table IV, 4.6.20, 4.6.21, 4.6.22.2, 4.6.23.2 and 4.6.24. The presence of any visual defect, any dimension not within the specified requirements or failure to pass any test shall be cause for rejection of the inspection comparison sample.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, the components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.1.1 Sandwich panels. Inspection shall be made of the sandwich panels specified in 3.4.1. Inspection records of all examinations and tests shall be maintained for each lot of panels and a certified copy of the inspection report for each production lot of sandwich panels shall be submitted to the contracting officer or his authorized representative. Each report shall verify that the inspections specified have been performed and complied with, shall include the actual test values obtained and the identity of the production lot represented by the tests performed. The inspection lot shall consist of one day's production not to exceed a maximum of 30 panels and the sample unit shall be one complete sandwich panel.

4.4.1.1.1 Visual examination. Sandwich panels shall be examined for visual defects. Any bulge, depression, dent or scratch shall be classified as a defect. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5.

4.4.1.1.2 Dimensional examination. Sandwich panels shall be examined for defects in dimensions. Any dimension not within the specified requirements shall be classified as a defect. The inspection level shall be S-3 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.4.1.1.3 Testing of sandwich panels. One panel from each production lot shall be tested as specified in table II for the characteristics shown, except as noted in table II. Results of each test shall be reported as pass or fail and failure of the sample panel to meet any of the specified requirements shall be cause for rejection of the lot.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:57Z
Check the source to verify that this is the currentversion before use.

TABLE II. Sandwich panel testing

Characteristic	Requirement paragraph	Test method or paragraph	Number of determinations per sample panel
Fire resistance 1/	3.4.1.1	4.6.1	1
Humidity 1/	3.4.1.1	4.6.2	1
Age 1/	3.4.1.1	4.6.3	1
Face sheet to core bond:			
Peel strength	3.4.1.1	4.6.4.1	2
Shear strength:	3.4.1.1	4.6.4.2	
TL direction			1
TW direction			1
Tension strength	3.4.1.1	4.6.4.3	2
Compressive strength	3.4.1.1	4.6.4.4	2
Continuity of bond	3.4.1.1	4.6.4.5	-

1/ Testing of the panels for fire resistance, humidity and age shall be performed on the initial lot of panels only.

4.4.1.1.4 Sandwich panel, with inserts. One complete sandwich panel shall be randomly selected from the initial production lot of panels. The inspection lot shall be of the size specified in 4.4.1.1. Four of each different size insert indicated in table I (total of 16 inserts) shall be potted in this panel using the same techniques to be used for fabricating production panels. These inserts shall be located a minimum of 6 inches (152 mm) in from panel edges and 6 inches (152 mm) on center from each other. Two identical inserts of each size shall be tested as specified in 4.6.22.1 with the other two identical inserts tested as specified in 4.6.23.1. If failure of one or more inserts occurs, a new panel, again chosen randomly from the same inspection lot shall have eight or sixteen inserts potted in, as applicable. This procedure shall be repeated until no failures occur. All failures occurring during this testing shall be recorded, including the size and diameter of insert(s) failing and the failing mode and furnished to the Government.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:57Z
Check the source to verify that this is the current version before use.

4.4.2 In-process inspection.

4.4.2.1 Process examination. Examination of the following fabricating processes to establish conformance with specified requirements shall be made:

a. That panel materials are as specified on the drawings.

b. That aluminum edge extrusions, phenolic closeouts, and inserts are cleaned and prepared prior to bonding for compliance with 3.5.6.

c. That sufficient adhesive is applied to raw edge of sandwich panel to completely fill exposed paper core cells for compliance with 3.5.6.

d. That both sides of aluminum edge extrusion legs, the faying surfaces of the extrusions and panel skins are coated with adhesive for compliance with 3.5.6.

e. That use of the adhesive for bonding of edge extrusions, closeouts and inserts is in accordance with the adhesive manufacturer's instructions for compliance with 3.5.6.

f. That steel components are cadmium plated, prior to painting, for compliance with 3.8.1.1.

g. That insert size, location and shape are as specified in 3.5.6.1.

h. That aluminum surfaces are chemically treated prior to painting, for compliance with 3.8.1.2.

i. That forming of metal parts, shearing and clipping of sheet metal, bolt and rivet holes are in compliance with 3.12. Whenever nonconformance is noted, correction shall be made to the items affected and the process.

4.4.3 End item inspection. The inspection lot shall consist of all expandable shelter units offered for inspection at one time, except as otherwise specified in 4.4.3.3.

4.4.3.1 Visual examination. The completely fabricated expandable shelter units shall be examined for defects in table III. The inspection level shall be II with an AQL of 6.5 for major defects and 10.0 for total major and minor defects combined, expressed in terms of defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:57Z
Check the source to verify that this is the current version before use.

TABLE III. Classification of defects for visual characteristics

Examine	Defect	Classification Major	Minor
Finish	Not as specified, not finished where required	X	
	Not adherent, e.g., blistered, peeled	X	
	Runs, wrinkles, streaks or areas of no film		X
	Finish not dry, i.e., wet or tacky to touch		X
	Scratch, gouge, abrasion exposing prime coat or bare metal		X
	Any area of rust	X	
Design	Any characteristic not in accordance with specified requirement	X	
Construction and workmanship	Component or part fractured, split, punctured, dented or malformed	X	
	Component missing, or will not operate as intended except as otherwise classified herein:		
	Component not properly assembled or secured	X	
	Any operation omitted or not properly performed	X	
	Any functioning component which is inoperative or requires abnormal force to operate	X	
	Burr, rough or sharp edge, or sliver	X	
	Gaskets torn or split	X	
	Gaskets or seals painted or otherwise coated where serviceability may be affected		X
	Visible gaps between aluminum faces or extruded edges or closeouts	X	
Welding and brazing (where required)	Not type specified, missing where required	X	
	Incomplete, burn through area, pits, crack or fracture, or otherwise not fused	X	
	Slag inclusion, undercut, not smooth and uniform, scale or flux deposit not removed		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:57Z
Check the source to verify that this is the current version before use.

TABLE III. Classification of defects for visual characteristics (cont'd)

Examine	Defect	Classification Major	Classification Minor
Metal fasteners, bolts, nuts, screws, studs	Not type specified, broken, stripped or loose	X	
	Lockwasher missing where required		X
Rivets	Not type specified	X	
	Not sufficiently peened; not drawn tight; excessively peened		X
Marking	Missing, incomplete, not legible, not specified type or size, misplaced		X
Instruction plate; nameplate	Incomplete, not legible	X	
Instruction manual	Incomplete, not legible	X	

4.4.3.2 Dimensional examination. The completely fabricated expandable shelter units shall be examined for compliance with dimensions specified. Any dimension not within the specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.4.3.3 Testing of the end item. Each expandable shelter shall be tested as specified in 4.6.4.5, 4.6.5, 4.6.6, 4.6.20 and 4.6.21. Any nonconformance shall be cause for rejection of the unit. In addition, one expandable shelter, selected at random from each lot produced that have passed the visual and dimensional examination, shall be tested as specified in table IV, except as noted in table IV. A lot for this inspection shall consist of 25 shelters or less. Any nonconformance shall be cause for rejection of the lot. Disposition of non-conforming lots will be conducted in accordance with MIL-STD-105.

TABLE IV. Testing of the end item

Characteristic	Requirement paragraph	Test method or paragraph
Blackout	3.5.3	4.6.7
Structural soundness: 1/ Towing	3.5.4	4.6.8
Transporting: 2/ Dolly set	3.11.1	4.6.9

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:57Z
Check the source to verify that this is the current version before use.

TABLE IV. Testing of the end item (cont'd)

Characteristic	Requirement paragraph	Test method or paragraph
Railroad humping 2/		4.6.10
Hoisting		4.6.11
Stacking		4.6.12
Environmental:		
Water leakage	3.5.5.1	4.6.13
Low temperature	3.5.5.2	4.6.14
Humidity	3.5.5.3	4.6.15
Low pressure	3.5.5.4	4.6.16
Solar load	3.5.5.5	4.6.17
Heat transfer	3.5.5.6	4.6.18
Electro-magnetic compatibility, fluorescent lights 2/	3.5.7	4.6.19

1/ Examination for panel delamination shall be conducted upon completion of each characteristic test and shall be performed as specified in 4.6.4.5. Since the presence of any delamination is cause for rejection of the unit, and the lot, any further testing required should be discontinued.

2/ Testing for transporting, railroad humping, and electro-magnetic compatability of fluorescent lighting shall be conducted only on the preproduction sample unit or initial unit of production, if no preproduction sample is required.

4.4.4 Packaging inspection. An examination shall be made to determine that packaging and marking as required by section 5 are complied with. Defects shall be as indicated in table V. The sample unit shall be one shelter fully prepared for delivery. The lot shall be all of the shelters offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 2.5, expressed in terms of defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:57Z
Check the source to verify that this is the current version before use.

TABLE V. Examination of preparation for delivery

Examine	Defect
Markings	Omitted; incorrect; illegible; of improper size, location or method of application.
Materials	Component missing or damaged or otherwise defective.
Workmanship	Inadequate application of components, such as items not boxed or secured, and openings not closed.

4.5 Certificate of examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.6 Test methods.

4.6.1 Fire, resistance, sandwich panels.

4.6.1.1 Test specimen. The test specimen shall be a minimum 4 inches (102 mm) by 8 inches (203 mm) piece of metallic-faced, paper honeycomb core sandwich panel.

4.6.1.2 Apparatus. Apparatus shall consist of a standard 9.5 mm barrel Bunsen or Tirrill burner fitted with a 48 ± 1 mm wide wing top and any suitable holding devices which will support the specimen horizontally and allow an unobstructed expanse a minimum of 6 inches (152 mm) of the specimen (a typical arrangement may consist of two laboratory ring stands with adjustable flat-surface clamps which may be locked in place by set screw lock nut).

4.6.1.3 Procedure. Place the specimen on its supports and at an appropriate level for the test. Adjust the burner with the wing top to provide a blue flame whose visible portion is 1-1/2 inches (38 mm) high with clearly defined inner cone 1/4 inch (6 mm) high. Place the burner under the specimen so that the flame impinges transversely across the 4 inch (102 mm) width of the specimen surface. After one minute, remove the flame. The specimen shall be judged "nonburning" if there is no evidence of ignition. If ignition should occur, note the time in seconds, when the flame extinguishes and when smoldering or glowing is no longer visible (see 3.4.1.1 and 4.4.1.1.3.)

4.6.2 Humidity, sandwich panels.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:57Z
Check the source to verify that this is the current version before use.

4.6.2.1 Test specimen. The test specimen shall be a minimum 12 inch by 12 inch (305 by 305 mm) piece of metallic-faced, paper honeycomb core sandwich panel.

4.6.2.2 Apparatus. Apparatus shall consist of a suitable test chamber capable of maintaining the relative humidity and temperature required for the test.

4.6.2.3 Procedure. The test specimen shall be placed in the test chamber and subjected to an environment of 97 percent ± 2 percent RH at a temperature of 120° ± 5°F (48.9° ± 3°C) for a period of 7 days. At the end of the period the specimen shall be examined for compliance with humidity requirements of 3.4.1.1.

4.6.3 Accelerated aging, sandwich panels.

4.6.3.1 Test specimen. The test specimen shall be a minimum 12 inch by 12 inch (305 by 305 mm) piece of metallic-faced, honeycomb paper core sandwich panel.

4.6.3.2 Apparatus. Apparatus shall consist of a suitable oven capable of maintaining the temperature required for the test.

4.6.3.3 Procedure. The test specimen shall be placed in the oven and subjected to a temperature of 200° ± 5°F (93.3° ± 3°C) for a period of 7 days. At the end of the period the specimen shall be examined for compliance with aging requirements of 3.4.1.1.

4.6.4 Face to core bond, sandwich panel.

4.6.4.1 Peel strength.

4.6.4.1.1 Test specimen. Each test specimen shall be 3 ± 0.01 inches (76.2 ± 0.3 mm) wide and approximately 12 inches (305 mm) long. The core ribbon direction shall be lengthwise of the specimen.

4.6.4.1.2 Apparatus. The apparatus shall be of the climbing drum, peel test type as described for the sandwich peel test specified in MIL-STD-401.

4.6.4.1.3 Procedure. The test procedure shall be as specified for sandwich peel test specified in MIL-STD-401.

4.6.4.1.4 Computation. Computation of test results, to determine compliance with peel strength requirements of 3.4.1.1, shall be as specified in MIL-STD-401 except the torque necessary to bend facing shall be determined and subtracted from the average torque needed to peel the sandwich.

4.6.4.2 Shear beam strength.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:57Z
Check the source to verify that this is the current version before use.

4.6.4.2.1 Test specimen. Each test specimen shall be 3 ± 0.01 inches (76.2 ± 0.3 mm) by 8 inches (203 mm) long with the core ribbon direction (L) lengthwise to the length of the panel, or transverse (W), as required.

4.6.4.2.2 Apparatus. Apparatus shall consist of three loading blocks; one loading block, 3 inches by 2 inches by 3/16 inch (76.2 by 50.8 by 1.6 mm) thick steel sheet and two reaction blocks 3 inches by 2 inches by 3/16 inch (76.2 by 50.8 by 1.6 mm) thick steel sheet; three self-aligning "V" blocks, and a testing machine capable of applying a load at a constant rate of movement.

4.6.4.2.3 Procedure. The test specimen shall be set up for center-point loading on a 6 inch (152 mm) span. The specimen shall be loaded in bending at a constant rate of movement of the movable head of the testing machine of not less than 0.025 (0.01) and not more than 0.050 inch per minute (0.02 mm/s) until failure occurs.

4.6.4.2.4 Computation. The face to core bond shear strengths to determine compliance with the requirements of 3.4.1.1 shall be calculated using the following formula:

$$F_s = \frac{L}{2\ (\text{specimen width})\ (\text{thickness of sandwich panel})}$$

where: F_s = shear stress (p.s.i.)

L = applied load

4.6.4.3 Tension strength, sandwich panel. Sandwich panel tension tests shall be performed as specified in MIL-STD-401 with specimen sizes of 2 inches by 2 inches (51 by 51 mm) and the results shall be examined for compliance with the tension requirements of 3.4.1.1.

4.6.4.4 Compression strength, sandwich panel.

4.6.4.4.1 Test specimen. Each test specimen shall be 4 inches by 4 inches (102 by 102 mm).

4.6.4.4.2 Apparatus. Apparatus shall consist of a testing machine that is capable of applying a load at a constant rate of movement.

4.6.4.4.3 Procedure. The specimen shall be loaded uniformly in flatwise compression over the entire surface of the specimen at a constant rate of .003 inch per minute (0.001 mm/s) ± 25 percent until failure occurs. The load shall be applied to the specimen through blocks of the self-aligning type and the test shall be performed at room temperature with dry specimens.

4.6.4.4.4 Computation. Flatwise compressive strength to determine compliance with the compression requirements of 3.4.1.1 shall be calculated as follows:

Source: https://assist.dla.mil -- Downloaded: 202640-21T22:57Z
Check the source to verify that this is the currenversion before use.

$C = P/A$

Where:

C = Compressive strength in psi (Pa)

P = Ultimate load in pounds (N)

A = Area of test specimen in square inches (m^2)

4.6.4.5 Continuity of bond, sandwich panel. Continuity of face to core bond shall be determined by tapping the face sheets over the bond areas with a coin or small plastic hammer to produce the sound effect for detecting a void area (see 3.4.1.1). A comparatively dull sound is indicative of a void area or delaminated area.

4.6.5 Operational test. The unit shall be placed on even ground and leveled through the use of the jacks. A complete cycle of leveling, opening and closing of the shelter shall be conducted to determine compliance with the operational requirements of 3.5.1.

4.6.6 Internal air leakage. With the shelter in the expanded condition and with all openings closed and the anti-backdraft damper taped or pegged closed, an air supply shall be applied to the unit and an internal pressure of 0.03 inch of water (water gauge) (74.7 Pa) shall be obtained and maintained. The pressure and flow shall be stabilized. Any additional air supplied to maintain the specified internal pressure shall be recorded to determine compliance with 3.5.2.

4.6.7 Blackout. The expanded unit shall be placed in a dark environment. A 40 watt fluorescent light shall be suspended from the ceiling brackets of the unit. With all covers in place and doors closed, and blackout curtains over all windows, observe the unit from a distance of 30 feet (9 m) to determine compliance with 3.5.3.

4.6.8 Towing. The unit, in its closed condition, and with an internal dummy load of 5500 pounds (2.5 Mg) shall be towed on its skids over rough terrain for a distance of 1000 feet (305 m) forward and 1000 feet (305 m) backward at a speed of 3 to 5 miles per hour (4.8 to 8.0 km/hr). At the end of the test the unit shall be examined for compliance with towing requirements of 3.5.4.

4.6.9 Transportation. The closed unit, with internal dummy load of 5500 pounds (2.5 Mg), shall be towed on a transporter dolly (see 6.5) over road surfaces in accordance with the roadability requirements listed below.

Type of surface and speed	Distance each test
Belgian block - 15 - 20 mph (24.1 to 32.2 km/hr)	50 miles (80.5 km)
Graded gravel road - five sudden stops from 15 mph (24.1 km/hr)	---

Source: https://assist.dla.mil -- Downloaded: 20261-21T22:57Z
Check the source to verify that this is the current version before use.

Two-inch washboard - 5 mph (8.0 km/hr)	4100 feet (1.2 km)
Two to four-inch radial washboard - 15 mph (24.1 km/hr)	1125 feet (0.3 km)
Three-inch spaced bumps - 20 mph (32.2 km/hr)	1300 feet (0.4 km)
Cross country No. 1 (moderate) - (speed shall vary from 5 to 20 mph (8.0 to 32.2 km/hr)	50 miles (80.5 km)

The unit shall be examined for compliance with transportability requirements of 3.5.4.

4.6.10 Railroad humping test. The shelter shall be in the folded position with an internal dummy load of 5500 pounds (2.5 Mg). The shelter unit shall be loaded in the manner normally used for shipment on a railroad flat car, (see Fig. 1). In addition, there shall be also required one standard railroad box car coupled to one standard railroad gondola car. The gondola shall be loaded with a minimum of 20 tons (18 Mg). These cars shall be equipped with standard draft gear coupling and the air brakes shall be set in the emergency application position on both cars. The test shall be conducted on a flat stretch or track. The test car, traveling at a minimum of 8-1/2 miles per hour (13.7 km/hr), shall be impacted against the two stationary cars. The test shall be repeated with the impact being made in the opposite direction. The shelter shall be examined to determine compliance with railroad humping requirements of 3.5.4.

4.6.11 Hoisting. The unit shall be tested with an internal dummy load of 25,000 pounds (11.3 Mg) minus the weight of the empty unit. The dummy load shall be evenly distributed over the entire floor area. The unit shall then be hoisted off the ground by means of a four-cable sling, having cable lengths such as to form 45 degree angles to the roof plane when attached to the lifting rings of the unit and the source used to lift the unit. The unit shall be held approximately 6 to 10 inches (152 to 254 mm) above the ground for a minimum of 30 minutes. At the end of the period, the unit, floor and rings shall be examined for compliance with hoisting requirements of 3.5.4.

4.6.12 Stacking test. The completely packed shelter (internal dummy load of 5500 pounds (2.5 Mg) in the folded condition shall be subjected to stacking. If two completely packed folded shelters are available, one shelter shall be placed upon the other and the two shall be oriented in the same position and remain stacked for 2 hours. If a second shelter is not available, a weight equivalent to a completely packed shelter shall be placed upon the shelter to be tested for a period of 2 hours. The weight shall be equally distributed and shall have a surface contact area which will simulate another shelter. After 2 hours, the shelter or weight shall be removed and the tested shelter examined to determine compliance with stacking requirements of 3.5.4.

Source: https://assist.dla.mil -- Downloaded: 202640-21T22:57Z
Check the source to verify that this is the currentversion before use.

4.6.13 Water leakage. The shelter shall be exposed to a simulated rainfall of 4 ± 1 inches (102 mm) per hour, as measured by a U.S. Weather Bureau type gauge. Direction of the rainfall shall be angled at 45 degrees from the vertical and the water pressure shall be not less than 45 psig (310 kPa). The roof with each of the major sides of the shelter in succession shall be exposed to this angled rainfall for periods of 30 minutes. The rainfall shall be dispersed uniformly, over the entire test area. This test shall be performed for 2 hours with the shelter in the closed configuration and for 2 hours with the shelter in the erected configuration. All openings, doors and panels shall be closed and latched. After completion of the test, the unit shall be examined for compliance with 3.5.5.1. Water leakage tests are to be conducted with adapter ducts (2 each) (P/N 5-4-2013) (81337) attached in erected position. Air duct assemblies are to be attached to adapters.

4.6.14 Low temperature. The completely packed folded shelter shall be placed in a cold chamber; the chamber temperature shall be lowered to -65° ± 5°F (-53.9 ± 3°C) and the shelter allowed to stand at this temperature for 24 hours. The complete shelter shall then be erected at this temperature and heated air shall be applied to the shelter air-conditioning supply inlet at the following conditions:

Temperature	250°F (121°C)
Air-flow	360 cfm (0.17 m^3/s)
Pressure	0.4 inch (water gauge) (99.5 Pa)

The temperature inside the shelter shall be maintained at 60°F (15.6°C) for a period of at least 4 days. The shelter shall be folded and packed while in the cold chamber. The folded unit shall be removed from the cold chamber. Immediately following removal, the shelter shall be unpacked and erected to determine compliance with the requirements of 3.5.5.2.

4.6.15 Humidity. The shelter shall be placed in the test chamber at ambient conditions in the closed position, with all covers in place, doors locked and the folding panels in the closed position. The temperature and relative humidity shall be raised gradually to 100° ± 5°F (37.8 ± 3°C) and 97 ± 2 percent RH and allowed to remain at these conditions for 24 hours. The shelter shall then be allowed to return to ambient conditions either within or upon removal from the test chamber. After 1 hour the shelter shall be examined to determine compliance with 3.5.5.3.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:57Z
Check the source to verify that this is the currentversion before use.

4.6.16 Low pressure (altitude). The closed shelter with all openings closed (except that the cap shall be removed from the tube above the access door) shall be placed in an altitude chamber at ambient conditions. Reduce the chamber internal pressure to 87.5 mm of mercury (11.7 kPa). Maintain the chamber pressure for a period of not less than 1 hour. After completion of the test and upon return to ambient conditions, the unit shall be examined to determine compliance with 3.5.5.4. As an option, individual sub-assembled panels equivalent to a complete unit may be tested in an autoclave in accordance with the above requirements. The Government representative shall randomly select each set of panels that will be tested.

4.6.17 Solar load test. The unit shall be placed in a chamber where the temperature can be stabilized at 100° ± 5°F (37.8 ± 3°C). An addition solar load sufficient to raise the outer skin temperature to 200° ± 5°F (93.3 ± 3°C) shall be applied to the fixed roof and one folding roof. The solar load shall be applied so that the 200°F (93.3°C) surface temperature is attained in 7 hours and maintained for an additional 4 hours. Instrumentation shall be provided so that at least five places can be measured on each surface (fixed roof and folding roof). At least three places on each surface (including the centers) shall read 200° ± 5°F (93.3° ± 3°C) during the test. The inside temperature of the unit, measured approximately 1 foot (.3 m) from the center of the panels being tested, shall be maintained at 75° ± 5°F (23.9 ± 3°C) during the entire period of the test. Upon completion of the solar load test, the outside skin temperature shall be reduced to 75° ± 5°F (23.9 ± 3°C) within 2 hours. Coin tapping tests shall be performed before and after the solar load test to determine compliance with 3.5.5.5

4.6.18 Heat leakage test for coefficient of heat transfer. A heat leakage test shall be conducted with the shelter erected in a controlled ambient temperature and a monitored heat input supplied to the shelter to determine compliance with 3.5.5.6. All calculations, using the formula below, and all data on time, temperature, and heat input shall be recorded.

$$\text{Heat leakage} = \frac{\text{Total watt-hours input x B.t.u/watt-hour}}{\text{Length of test (hours) x temperature difference (°F) x total outside area of shelter } (ft^2)}$$

The temperature difference shall be the difference between average shelter temperature and average ambient temperature. The temperature difference shall be not less than 60°F (33.3°C) nor more than 80°F (44.4°C). The ambient temperature shall be the average of eight readings taken at points located 6 inches (152 mm) from each outside corner of the shelter. The internal, or shelter temperature, shall be the average of eight readings taken at points located 6 inches (152 mm) from the inner walls at each corner of the shelters. Eight ambient and shelter temperature readings shall be taken at consecutive 30 minute intervals with ambient temperature variations not more than plus or minus 2°F (1.0°C) and shelter temperature variations not more than plus or minus 1°F (0.5°C). At least one interior and one exterior temperature reading shall be taken on a recording instrument.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:57Z
Check the source to verify that this is the currentversion before use.

4.6.19 Electromagnetic compatability. The fluorescent light fixtures of the preproduction sample or initial unit of production, as applicable, shall be tested by the contractor in accordance with test methods CE03 and RE02 of MIL-STD-462. Results shall be examined for compliance with the electromagnetic compatibility requirements of 3.5.7. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The contractor shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Test Report and the requirements meet MIL-STD-462.

4.6.20 Electrical system test. To determine compliance with the continuity and operational requirements of 3.5.7, the following test procedure shall be performed. Failure to comply with any of the requirements specified shall be cause for rejection. The electrical sources listed in table VI are necessary to perform the tests.

TABLE VI. Test electrical sources

Item	Description	Tolerance	Remarks
Electrical power	External, 28 V	± 10%	Required to check out emergency lighting system
Electrical power, 400-cycle (Hz)	4-wire service 120/208 V 5 kW	± 10%	Required to test receptacles, lights, and heaters
Electrical power, 60-cycle (Hz)	3-wire service 120/240 V (no power requirements)	± 10%	Required to test receptacles in 60-cycle service

Precautionary notes:

1. Prior to connecting external source of energy, disconnect all circuit breakers. Close breakers as directed.

2. When checking sump float level switch, insure pump does not operate with sump empty.

Connect 400 (Hz) power input cable to 400 Hz shelter input connector located in the service entrance box on outside of shelter. Close circuit breaker to external power source and check voltages at output connectors in the service entrance box as follows: Voltages at pins of 400 Hz connector J-2 should read 120 V between pin D (neutral) and pins A, B and C. No voltage should be read on pin E. Voltage between pins A-B, A-C, or B-C should read 208 V. No voltage should be present between any pins in 60 Hz connector J-4.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:57Z
Check the source to verify that this is the currentversion before use.

Connect 60 Hz power input cable to 60 Hz shelter input connector located in the service entrance box on outside of shelter. Close circuit breaker to external 60 Hz power source and check voltages at output connectors in the service entrance box as follows: Voltage at pins of the 60 Hz connector J-4 should read 120 V from pin C (common) to pins B and D. Voltage between pins B and D shall be 240 V.

The surgery light external power receptacle J-17 shall be checked as follows: With CB-9 open, check continuity between ground and pin E. Close circuit breaker CB-9 (400 Hz transformer surgery light). The voltage present between pins C and D of the surgery light external power receptacles should measure 120 V a.c. Circuitry for surgical light receptacle may be checked by jumping pins A and B at receptacle J-17 and reading continuity between pins A-1 and A-3 at receptacles J-18, J-19 and J-20 when toggle switches S-8, S-9 and S-10 are placed in the "ON" position or by utilizing an external 28 V d.c. power source connected to pins A and B of J-17 and checking surgery light receptacles J-18, J-19 and J-20 for 28 V d.c., between pins A-1 and A-3 when surgery light toggle switches S-8, S-9 and S-10 are in the "ON" position.

Close circuit breaker CB-8 (area lights), operate toggle switches S-1 and S-2 (wall switches marked ceiling light switches). With the fluorescent fixture connectors P-21 and P-22 plugged into the receptacles J-21 and J-22 in ceiling outlet boxes, the ceiling lights should light. In the event a light fails to light check for the presence of 120 V a.c. between pins X and Y at the ceiling receptacle for that light to establish whether lamp failure has occurred. Pin W is common to ground.

Close circuit breakers CB-4 (Receptacles, R.H. Ceiling) and CB-13 (Receptacles, L.H. Ceiling) and measure 120 V, 60 Hz output power at ceiling receptacles. This check can be performed by using a 120 V lamp connected to a standard twistlock cap inserted into the various outlets while observing the lamp for indication of power. A similar check of the 60 Hz 120 V wall receptacles shall be made by closing CB-5 (Receptacle R.H. Wall) and CB-12 (Receptacle L.H. Wall) and of the 400 Hz, 120 V wall receptacles, by closing CB-6 (Receptacle L.H. Wall) and CB-7 (Receptacle R.H. Wall).

Check of the 400 Hz airlock receptacle shall be made by closing CB-2 and measuring 210 V a.c. across pins C-D of receptacles J-13 and J-14. Pin E shall indicate continuity to ground.

Close CB-17 (Heater, Water Line) and CB-18 (Heater, Water Hose) and disconnect P-33 and P-34 located in the water service entrance box. Measure 120 V, 400 Hz power between pins C and D of receptacle J-33 and J-34. Reconnect P-33 and P-34 and provide a cooling source less than 38°F (3.3°C) to lower temperature of thermoswitches S-4, S-5 and S-7. Thermoswitches should activate when the temperature drops to 40° $\pm$ 5°F (4.4° $\pm$ 3°C). While the thermoswitches are activated check strip heaters with a contact probe or equivalent and observe and record stabilized heater temperatures. Remove the cooling source and the thermoswitches should deactivate when a temperature of 60° $\pm$ 5°F (15.6° + 3°C) is reached.

Source: https://assist.dla.mil -- Downloaded: 202640-21T22:57Z
Check the source to verify that this is the current version before use.

Close circuit breakers CB-11 (Sterilizer #1) and CB-14 (Sterilizer #2). 400 Hz power should be present at receptacles J-5 and J-6. Pins A, B and C of J-5 or J-6 are connected to the three phases and pin D is connected to neutral. With a voltmeter measure 208 V between phases of 120 V between any phase and neutral.

Close circuit breaker CB-3 (Receptacle, X-Ray). At receptacle J-7 measure 240 V between receptacle contacts A and C, and 120 V between A-B and C-B. Pin D shall show continuity to ground.

Perform check of humidistat circuitry as follows: With CB-10 open, install jumper between pins A and B of receptacle J-36 and G and F of J-31. Close CB-10 and measure 120 V between pins A-B, B-D, and C-D and 208 V between A-B, B-C, and C-A. Pin E should indicate continuity to ground.

Check external water hose heater receptacles by closing CB-18 and CB-19 and measuring 120 V between pins C and D at receptacles J-15 and J-16.

To check the operation of the sump pump, remove sump cover and add water. Close pump circuit breaker (CB-16) and check proper operation of the float level switch in the following manner: Add water to the sump until the upper float energizes the pump, 7 $\pm$ 1/2 inches (178 $\pm$ 13 mm). The pump should continue to operate until the level has been lowered sufficiently to deactivate the circuit, 2 $\pm$ 1/2 inches (51 $\pm$ 13 mm). (Observe precautionary note).

Connect an ohmmeter between the two terminals on the telephone jack located inside the unit. The resistance reading should show infinity. Connect the two terminals on the telephone jack located in the power input panel. Read continuity across the two terminals of the telephone jack located inside the shelter. Remove the connection between the two terminals on the jack in the power input panel.

When a Utility Element with external 28 V d.c. power supply is not available the 28 V emergency lighting system may be tested as follows: Remove external 400 and 60 Hz power cables and connect 28 V d.c. at pins A (Positive d.c.) of the 60 Hz output plug, and pin E (Negative d.c.) of the 400 Hz output plug. The light should function in conjunction with either wall light switch.

4.6.21 Plumbing system. A water supply with a pressure of 60 pounds per square inch psig (0.4 MPa) or an air supply with a pressure of 60 psig (0.4 MPa) shall be available for testing the plumbing system. Each line shall be subjected to 60 psig (0.4 MPa) water/air pressure. All connections and the piping shall be examined while under pressure for compliance with 3.5.8. Vacuum piping will be subjected to an air test. All lines hydrostatically tested will be purged of water at completion of test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:57Z
Check the source to verify that this is the current version before use.

NOTE: Should an air source be elected, a valve shall be installed upstream of a pressure gage in order to make possible disconnection of the air source while monitoring for a pressure drop over a period of 5 minutes. No drop is permissible. An accumulator tank of up to 25 gallons (94.6 Liters) is acceptable. The drain system does not require pressure testing.

4.6.22 Insert torque.

4.6.22.1 Sample panel torque. Torque specified in table I shall be applied to two identical inserts of each size which have been potted in the sample panel described in 4.4.1.1.4 to determine compliance with 3.5.6.1. The torques shall be applied using a torque wrench and hex head screws of a steel alloy which will allow attainment of specified torques without screw failure.

4.6.22.2 Proof torque. A proof torque equal to 80 percent of the applicable torque in table I shall be applied to randomly selected inserts of each different size on each panel prior to final assembly to determine compliance with the requirements specified in 3.5.6.1. The number of inserts, torqued for each size, shall be in accordance with MIL-STD-105. Additionally all inserts which are installed after final unit assembly, for attachment of lifting ring assemblies, must also be subjected to the torque test.

4.6.23 Insert pull-out.

4.6.23.1 Sample panel pull-out. Tensile pull-out loads specified in table I shall be applied to two identical inserts of each size which have been potted on the sample panel described in 4.4.1.1.4 to determine compliance with 3.5.6.1.

4.6.23.2 Proof pull-out. A proof tensile pull-out load equal to 80 percent of the applicable pull-out specified in table I shall be applied to randomly selected inserts of each different size on each panel prior to final assembly to determine compliance with the requirements specified in 3.5.6.1. The number of inserts, for each size, shall be in accordance with MIL-STD-105.

4.6.24 Durability test. One completely fabricated shelter shall be tested for conformance with 3.7. The test shall consist of 6 test cycles of 240 hours each for a total durability test time of 1440 hours and any failure as defined below shall consitute failure of the test. Prior to initiation of the test, The complete shelter shall be unpacked and repacked with a 5500 pound (2.5 Mg) dummy load uniformly distributed on the fixed floor.

a. The test cycle for the expandable shelter shall consist of:

(1) Load the shelter on a transporter dolly set.

(2) Transport the shelter (in its folded configuration) by transporter dolly set to an erection site a distance of 25 miles over cross-country No. 1 (moderate) terrain.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:57Z
Check the source to verify that this is the current version before use.

(3) Remove the shelter from the dolly set.

(4) Erect the shelter and redistribute the dummy load by uniformly distributing 1000 pounds (0.45 Mg) on each expanded floor and the remaining 3500 pounds (1.6 Mg) uniformly distributed on the fixed floor. Perform the internal air leakage test (4.6.6), electrical system test (4.6.20), and the plumbing system test (4.6.21). The shelter shall remain erect in the operational mode between erection and striking.

(5) Relocate the dummy load from the expanded floors to the fixed floor and strike the shelter.

b. A failure is defined as any malfunction which the operator cannot remedy by adjustment, repair or replacement action within 60 minutes using tools and parts normally available and which cause or may cause:

(1) Inability to commence operation.

(2) Cessation of operation during the cycle of 240 hours.

(3) Reduction of performance below an acceptable level, such as but not limited to: loss of electrical continuity; inability of any light to continue operation; or any plumbing that exhibits leakage or seepage.

(4) Personnel safety hazard by continued operation.

(5) Serious damage to the shelter or element thereof by continued operation.

Any malfunction attributed to operator error shall not be considered an accountable malfunction.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A.

* 5.1.1 Level A. The air ducts, air duct adapters, metallic tread plate, power cables, hoses, bellows, air distribution plenum and lighting assembly for airlock shall be packaged in commercial fiberboard boxes closed with tape. The airlock shall be compactly folded and secured. The fiberboard boxes, airlock and airlock adapter shall be secured within the shelter by means of the cargo net tie-down system. The instruction manual (see 6.4) shall be placed in a kraft paper envelope and together with the special tools, set in positions provided in the shelter and secured in such a manner as to be immobilized while in transit. All windows, air-conditioning vents, doors, and other openings shall be closed, covered, and secured with the means provided.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:57Z
Check the source to verify that this is the current version before use.

5.2 Packing. Packing shall not be required.

5.3 Marking. In addition to any special marking required by the contract or order, each shelter shall be marked in accordance with MIL-STD-129, with marking applied to tags as specified therein. Level of protection marking shall be A/A.

6. NOTES

6.1 Intended use. The expandable shelter covered by this specification is intended for use as an element of the MUST (medical unit, self-contained, transportable) System. The shelter is intended to serve as a universal element, suitable for surgical use, as a laboratory or pharmacy, for central material supplies, X-ray facility, or kitchen facility.

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.3, 4.3 and 6.3).
(c) When an initial production testing sample is required (see 3.3.1, 4.3.1 and 6.3.1).
(d) When an inspection comparison testing sample is required (see 3.3.2, 4.3.2 and 6.3.2).
(e) When airlock adapters, external power cables, cargo tie-down assemblies, bellows and airlock assemblies are to be Government furnished (see 3.10.1, 3.10.2, 3.10.3, 3.10.4, and 3.10.5).
(f) When engineering design conference is required (see 6.6).

6.3 First article. When a first article is required, it shall be inspected and approved under appropriate provisions of ASPR 7-104.55. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.3.1 Initial production test. When an initial production test sample is required, the contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the initial production test sample.

6.3.2 Inspection comparison test. When a comparison test sample is required, the contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the comparison test sample.

* 6.4 Manuals. Any requirements for equipment and instruction manuals for the expandable shelter covered by this specification should be included in DD Form 1423, Contract Data Requirements List and cited in the contract. Unless otherwise specified, no end item shall be shipped minus operator manuals without approval of the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 20260-21T22:57Z
Check the source to verify that this is the current version before use.

6.5 Transporting and railroad humping tests. The contracting officer should make all necessary arrangements for testing at a test site sufficiently in advance to minimize the possibility of any delay in testing after the scheduled arrival of the test unit. Equipment required for the transporting test is a standard 2-1/2 or 5-ton truck, and a Dolly Set, Lift, Transportable Shelter: Army (19208) 8736772, Model M689E1, (NSN 2330-00-937-1175).

6.6 Engineering design conference. When specified (see 6.2), within 10 days after award of contract and prior to any work on the preproduction sample, the contracting officer shall arrange an engineering design conference to discuss any questions concerning the details of Government drawings, instructions, and technical requirements of this specification. This conference is to assist the contractor in interpretation of the specification, and agreements made shall neither obligate the Government to accept equipment that is not in accordance with this specification nor relieve the contractor of his obligation to supply the equipment specified.

* 6.7 Sandwich panel review. The purpose of the requirements in paragraph 3.1 is to assist the Government in determining the capability of contractors to manufacture sandwich panels meeting all the requirements of this specification. Notice of completion of review of the process specification, laboratory report and test sample under these requirements shall in no way be construed as relieving a contractor from subsequently meeting all the requirements of this specification for the sandwich panels and the end item.

6.8 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

* 6.9 Government furnished property. The contracting officer should arrange to furnish the property listed in 3.10.

* 6.10 Government loaned property. The contracting officer should arrange to loan the property listed in 3.11.

6.11 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

Army - GL

Review activity:

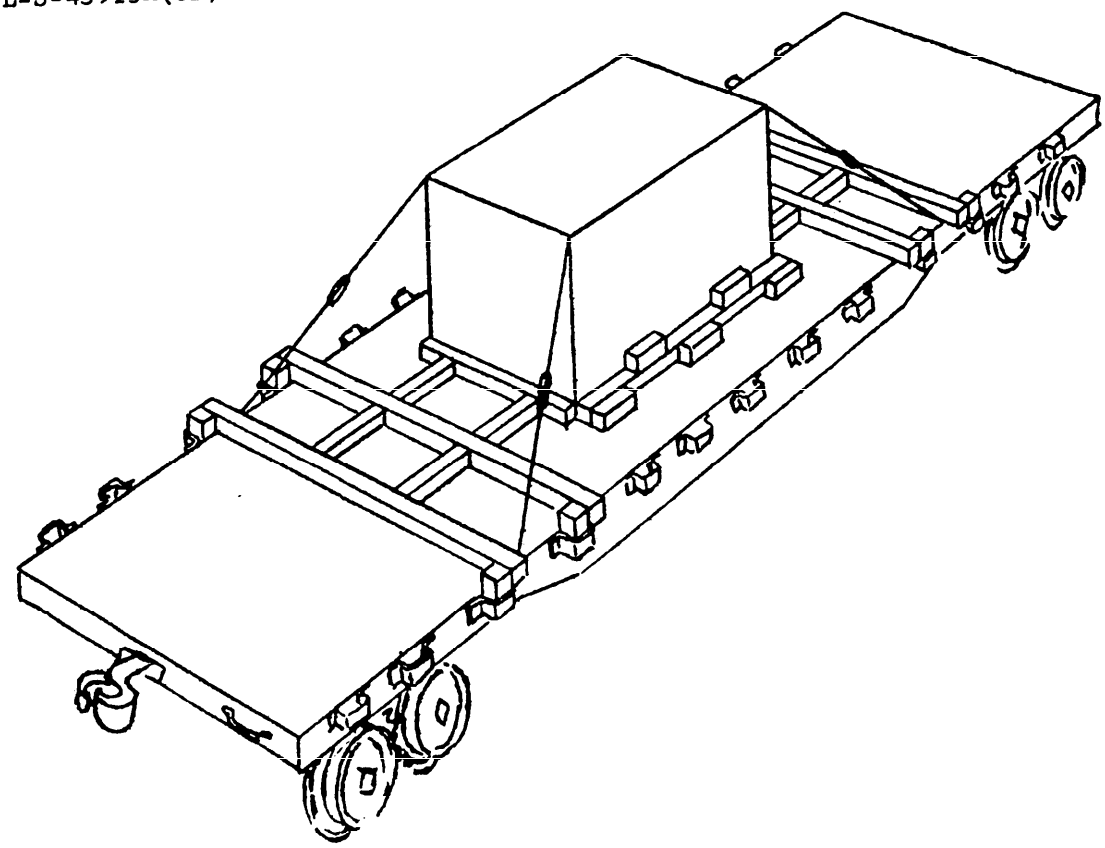
Army - MD

Preparing activity:

Army - GL

Project No. 5410-A216

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:57Z
Check the source to verify that this is the current version before use.



NOTES:

1. SAWED NOTCHES REQUIRED IN BLOCKING USED ALONG DOOR END & FRONT END TO CLEAR MOBILIZER BRACKETS TO OBTAIN TIGHT FIT ALONG FULL WIDTH.
2. TIE DOWNS TO CONSIST OF WIRE ROPE, TURNBUCKLES & FASTENERS.
3. TIMBER SHALL BE NEW & REVEAL NO INDICATION OF STRUCTURAL WEAKNESS.
4. ALL BLOCKING AND BRACING TO BE 8" × 8".

FIG 1 SHELTER(MUST) SECURED ON FLATCAR 5-4-1354 B

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:57Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER (Number) AND TITLE
MIL-S-43915A(GL) Shelter, Expandable For Medical Unit Self-Contained, Transportable (MUST)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 OCT 76 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:57Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
US ARMY NATICK RESEARCH and DEVELOPMENT COMMAND
NATICK, MASSACHUSETTS 01760

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD-314

U.S. MAIL

Commander
US Army Natick Research and Development Command
ATTN: DRXNM-TDI
Natick, MA 01760

NATICK OP 1 JUL 77

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:57Z
Check the source to verify that this is the current version before use.